



TM Forum Technical Specification

Implementation and Deployment Blueprints for Hybrid Environments

TMF070

Release 17.5.1

April 2018

Latest Update: TM Forum Release 17.5.1	TM Forum Approved
Version 0.4.2	IPR Mode: RAND

Notice

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Executive Summary

The TMF070 series of documents provides implementation and deployment blueprint for various types of platforms. The blueprints are based on the platform architecture and principles in TR262 and make extensive use of the TM Forum Open APIs.

This document (TMF070) provides a brief overview of each document in the associated series, and a summary of the various platforms defined in the series.

1. Introduction

The TMF070 series of documents provides implementation and deployment blueprint for various types of platforms. The blueprints are based on the platform architecture and principles in TR262 and make extensive use of the TM Forum Open APIs.

For each platform, requirements are provided. The required capabilities are mapped to TM Forum Open APIs (gaps are identified). There are also several detailed scenarios (down to the level of API operations) concerning management processes such as Resource Function (RF) Lifecycle Management, Product Pre-Ordering and SLA Violation Correction.

2. TMF070A

TMF070A is a detailed blueprint concerning a platform for managing multi-vendor hybrid (virtual and physical) infrastructures (and is included in the Agile OSS toolkit). The blueprint brings together multiple elements into one set of tools. This blueprint goes beyond the very basic management capabilities typically deployed with NFV today, and includes items such as:

- Resource Function Activation and Configuration (where Resource Function includes as special cases the ETSI NFV concepts of VNF and Network Service, and the IETF / SDN concepts of Service Function and Service Function Chain)
- Inventory Management
- Catalog Management
- Assurance.

Together these key capabilities begin to unlock the latent potential of the new technology in a standardized interoperable way, and allows realization of the long-awaited business benefits.

The TR225 series concerning resource function activation and configuration and the associated API in TMF664 are key parts of the blueprint. This API supports configuration and lifecycle management of resource functions (virtual, physical and mixtures, i.e., hybrid). Further, the TM Forum Information Framework (SID) has been updated to support resource functions and their associated specifications (i.e., descriptors). Together, these assets define a standardized interface that allows resource functions to be dynamically configured, which is a key capability to enable the dynamic network behavior that creates some of the most significant business benefits of virtualization.

3. TMF070B

TMF070B describes implementation and deployment platform blueprints beyond the one described in TMF070A (i.e., HIP). The document covers enhancements to the HIP described in TMF070A, HIP variations (e.g., Customer Facing HIP) and additional platforms at the service, product and customer layers.

TMF070B employs the same methodology as TMF070A, i.e., makes use of the platform architecture and principles in TR262 and many of the TM Forum Open APIs.

4. TMF070C

TMF070C provides descriptions of the relationships between the platform examples in TMF070A and TMF070B, and external architectures such as those in ETSI NFV, SDN and associated open source projects such as ONAP.

5. Platform Summary

The following table summarizes the platforms defined in TMF070A and TMF070B.

Platform	Layer	Description	Comment
Hybrid Infrastructure Platform (HIP)	Resource	Management of resources (virtual, physical and compound mixtures) HIPs can be layered in tiers CF HIP and SMP are consumers of HIP functional capabilities	This is sometimes qualified as Resource Facing HIP (RF HIP) to distinguish from the CF HIP
Customer Facing HIP (CF HIP)	Resource, Product and Customer	This platform is basically an encapsulation of HIP with added product and customer capabilities to expose resources directly to external consumers A CF HIP can directly access resources or can be layered on top of a separate HIP	CF HIP is designed not to support or make use of a service layer
Service Management Platform (SMP)	Service	This platform manages services. SMPs are designed to be consumers of HIPs.	
Customer and Product Management Platform (CPMP)	Product and Customer	This platform manages products and customers. CPMPs are designed to be consumers of SMPs. CPMP are not designed to directly access HIPs.	While not yet covered in TMF070B, it is possible to define separate platforms for Product and Customer.

6. Administrative Appendix

This Appendix provides additional background material about the TM Forum and this document. In general, sections may be included or omitted as desired; however, a Document History must always be included.

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.1.0	08-Jun-2017	Steve Fratini	Initial draft document
0.2.0	06-Jul-2017	Alan Pope	Added Administrative Appendix
0.3.0	06-Jul-2017	Alicja Kawecki	Minor formatting/style edits prior to publication for Fx17
0.3.1	14-Nov-2017	Adrienne Walcott	Minor cosmetic edits prior to publishing
0.4.0	22 Nov 2017	Stephen Fratini	Extensive revision for Release 2 of document
0.4.1	01 Jan 2018	Adrienne Walcott	Formatting/style edits prior to publishing
0.4.2	05 Apr 2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

6.1.2. Release History

Release Number	Date Modified	Modified by:	Description of changes
17.0.0	08-Jul-2017	Alan Pope	Final document
17.0.1	14-Nov-2017	Adrienne Walcott	Updated to reflect TM Forum Approved Status
17.5.0	22 Nov 2017	Stephen Fratini	Initial draft of Release 2

Release Number	Date Modified	Modified by:	Description of changes
17.5.1	05 Apr 2018	Adrienne Walcott	Updated to reflect TM Forum Approved Status

6.2. Acknowledgments

This document was prepared by the members of the TM Forum ZOOM team:

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